

Secure Sense: Employee Behavior Pattern Recognition and Recommendation System

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Abstract— Secure Sense is a comprehensive system designed to analyze employee behavior patterns and enhance security measures within an organization. By utilizing data on employees' login/logout activities and URL logins, the system provides insights into behavior trends and potential security threats. Additionally, Secure Sense incorporates a content-based recommender system to suggest similar files to users based on their behavior and preferences. This recommender system leverages machine learning algorithms to calculate cosine similarity between file contents and employs K-means clustering to categorize files. By recommending top files from clusters that users have not yet accessed, Secure Sense aims to boost productivity and ensure that employees have access to relevant resources, thereby improving both security and efficiency within the organization

I. INTRODUCTION

In this digital age, information security is the foundation upon which all organizations stand. However, traditional security strategies often fall short in detecting complex threats and insider attacks [1]. Secure Sense addresses these challenges by analyzing employee behavior patterns through ongoing monitoring. This approach enables organizations to identify atypical activities early, strengthening their defenses and preparing more effectively by gaining advanced knowledge of potential risks. By leveraging the strengths of machine learning algorithms, Secure Sense provides an advanced yet user-friendly tool designed to enhance data protection protocols and safeguard sensitive information [2]. Additionally, the system incorporates a content-based recommender system that analyzes employee behavior to suggest relevant content files, thereby improving efficiency and ensuring users have access to the most pertinent information [3]. This innovative method ensures that organizations can proactively identify and mitigate security threats while simultaneously enhancing data accessibility and organizational resilience [4].

II RELATED WORK

Problem Statement

The current field of security primarily relies on reactive measures, such as scheduled audits and incident response, which, although effective, are often time-consuming and lag behind emerging threats [1].

Secure Sense addresses these limitations by analyzing employee behavior patterns and building a content-based recommender system to enhance both security and productivity. By continuously monitoring user activities, Secure Sense can identify unusual behavior early, allowing companies to take proactive measures against potential threats [2]. Additionally, the system improves user efficiency by recommending relevant content files based on behavior analysis, ensuring that employees have access to the most pertinent information [3]. This dual approach not only bolsters security by preemptively identifying risks but also enhances overall organizational resilience and data accessibility [4,5].

A. Background Work

Traditional security strategies often struggle to detect complex threats and insider attacks effectively. Secure Sense addresses this challenge by employing machine learning algorithms to analyze employee behavior patterns through continuous monitoring. This approach enables early identification of atypical activities, strengthening organizational defenses and facilitating proactive risk mitigation [6]. Secure Sense leverages unsupervised learning techniques like clustering and anomaly detection to identify unusual behavior patterns in employees' file interactions [7]. These methods can uncover potential security risks or policy violations, enhancing data protection protocols and safeguarding sensitive information [8]. Furthermore, Secure Sense incorporates a content-based recommender system that analyzes employee behavior and file content to suggest relevant files [9]. This feature improves efficiency by ensuring employees have access to the most pertinent information, ultimately enhancing organizational resilience and data accessibility. By combining advanced machine learning techniques for behavior analysis, anomaly detection, and content-based recommendations, Secure Sense provides a comprehensive solution for organizations to proactively identify and mitigate security threats while optimizing information accessibility.

III. SYSTEM DESIGN & IMPLEMENTATION DETAILS

Secure Sense adopts different types of machine learning algorithm such as Isolation Forests to carry out employee's behavior analysis. This system takes data on login/logout events, URL access, and other events to preprocess it and obtain the features. The features extracted here are url length. These features are employed to enable machine learning systems for fault detection and pattern recognition. Likewise, the system may recognize patterns as browsing or accessing wrong files, amount of productive work, and site connection to malicious URLs. These data consist of comprehensive HTTP, file, logon, login, and device data provide a clear view of the employee behavior to improve the security as well as operational efficiency.

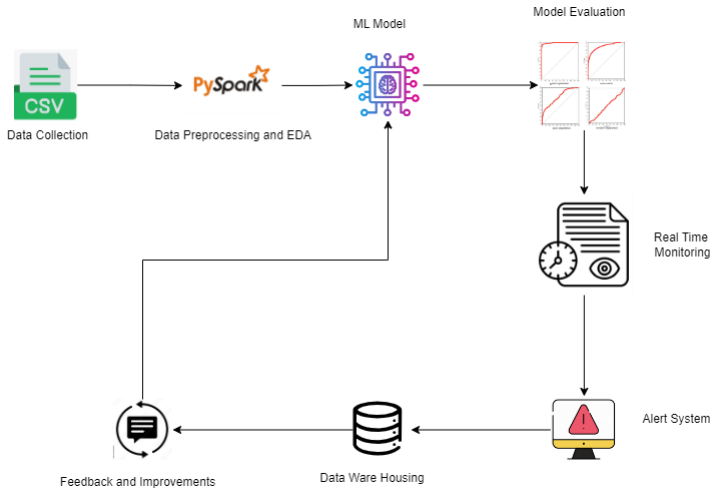


Fig 1: System Architecture

1.Data Collection: The process starts with data collection, represented by the CSV file icon, which suggests that data is collected and stored in a CSV (Comma-Separated Values) format.

2.Data Preprocessing and EDA: The collected data is then pre-processed and subjected to Exploratory Data Analysis (EDA) using PySpark, a popular distributed computing framework for big data processing.

3.ML Model: The pre-processed data is used to train and evaluate an ML model, depicted by the cog-like icon.

4.Model Evaluation: The trained ML model is evaluated using various techniques, as indicated by the visual representations (scatter plots) under "Model Evaluation."

5.Real-Time Monitoring: The trained and evaluated ML model is then deployed for real-time monitoring, where it continuously processes incoming data and generates predictions or analyses patterns.

6.Alert System: The real-time monitoring system is integrated with an alert system, represented by the exclamation mark icon, which triggers alerts or notifications based on predefined thresholds or anomalies detected in the data.

7.Data Ware Housing: The data, including the collected raw data, pre-processed data, and model predictions, is stored in a data warehouse, represented by the database icon.

8.Feedback and Improvements: The system incorporates a feedback loop, indicated by the feedback icon, which allows for model improvements based on the real-time monitoring results, user feedback, or other relevant inputs.

IV. DATA HANDLING & EDA

Secure Sense project data management comprises data gathering, data preprocessing, and data analysis of large amounts of the employee behavior data and getting relevant insights. The system accumulates data from different sources, like HTTP logs, file access logs, login/logout events, device info, and so forth. The data that we are going to work consists of multiple files like:

1. Login_info.csv (34.2 MB)
2. device_info.csv (17.9 MB)
3. Files_info.csv (108 MB)
4. User_info.csv (10 MB)
5. Http_Information.csv (5.11 GB)

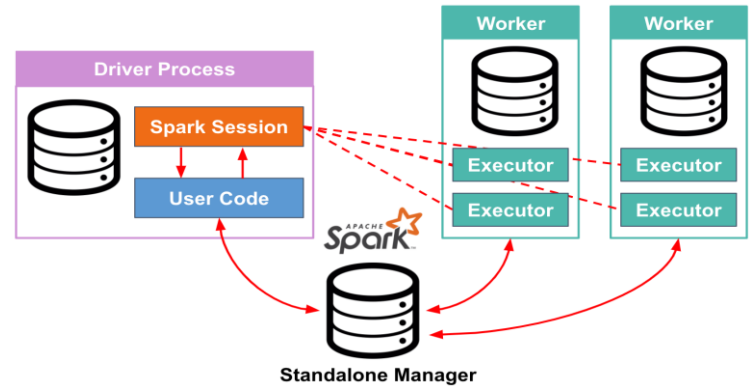


Fig 2: Pyspark Architecture

We utilized PySpark for handling these large datasets. PySpark is the Python API for Apache Spark, a powerful distributed computing system designed for big data processing and analytics. PySpark is particularly useful for handling big data due to its ability to distribute data processing tasks across a cluster of computers, enabling parallel execution and efficient management of large datasets that are too big for a single machine's memory. It supports a variety of data formats and integrates seamlessly with Hadoop, allowing it to read from and write to HDFS. In our project, PySpark was instrumental in managing and analyzing the extensive employee behavior logs, URL access patterns, and content files. We leveraged PySpark's Data Frame API

for data manipulation, SQL-like queries, and joining large datasets, as well as its machine learning library, MLlib, for scalable implementations of clustering and anomaly detection algorithms. This allowed us to perform complex operations like TF-IDF calculation and sentiment analysis at scale, ensuring efficient and timely analysis of the vast amounts of data involved in our project

Preprocessing plays a crucial role in preparing the data for analysis. It includes steps such as data cleaning, feature extraction, and normalization. Data cleaning involves handling missing values, removing duplicates, and ensuring data consistency. Feature extraction involves selecting relevant features from the raw data that can be used by machine learning models for anomaly detection and pattern recognition. Normalization is performed to scale the data to a common range, ensuring that all features contribute equally to the analysis.

A. logon_Info.csv:

This file consists data like id, pc, date, user and activity (logon, /logoff). After observing the data, we found that there are missing values in the activity column as few entries has logon time with missing logon time while few others have vice-versa. The rows which have both logon and logoff times we tried to calculate the number of predictive hours per user. For the missing values we thought of handling them using collaborative filtering like we would like to predict the logoff/logon time by using similar user logoff/logon pattern.

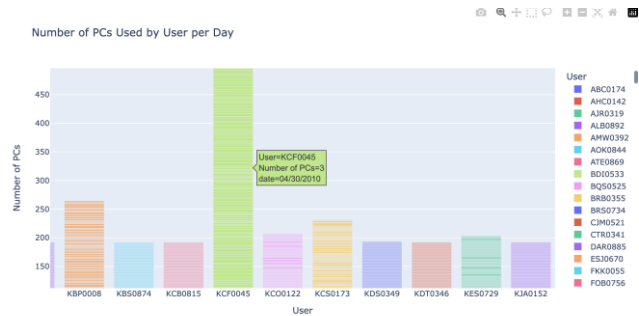


Fig 3: Number of Pc’s used by user per day

Additionally, our exploration of the temporal aspects of user activity, depicted in the "Clustering on Users log in and Logoff Timings" scatter plot, unveiled distinct clusters of login and logoff events. These clusters signify different patterns of activity among users, ranging from consistent login and logoff timings to irregular patterns, which may reflect variations in work habits or preferences. These insights gleaned from our exploratory analysis not only provide a comprehensive understanding of user behavior but also lay the groundwork for further research and the development of tailored strategies to optimize system resources, enhance user experience, and strengthen security measures.



Fig 4: Clusters on users Login and Logoff timings

B. http_info.csv:

This file provides the information about URLs accessed per user on what date, from which device and also the content present in URL. As the data was in good form for processing to ML model for clustering the URLs as spam/ham, there wasn’t, any need for pre-processing the data. I have used tokenizer for converting URL into words or tokens, also used Count Vectorizer which Converts a collection of text documents (in this case, the tokens) into a matrix of token counts. This helps in quantifying the frequency of each word post that used IDF (Inverse Document Frequency); Down-weights tokens that appear frequently across documents. This helps in giving more weight to tokens that are more specific to individual URLs. With these features that have performed K-MEANS clustering to split the URLs into two groups/cluster. I would like to experiment performing clustering using different features like considering URL’s length or domain specific words and also try different algorithm like hierarchical clustering, Gaussian Mixture Models (GMM).

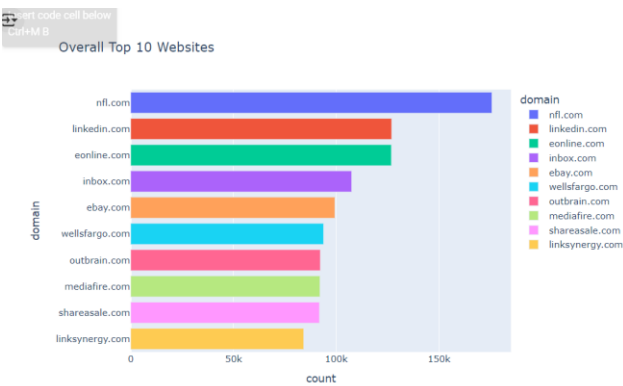


Fig 5: Overall Top 10 Websites

year	month	url	count
2010	1	http://nfl.com/Gr...	6244
2010	1	http://eonline.co...	4082
2010	1	http://linkedin.c...	4048
2010	1	http://inbox.com/...	3676
2010	1	http://ebay.com/S...	3516
2010	1	http://outbrain.c...	3194
2010	1	http://wellsfargo...	3183
2010	1	http://shareasale...	3084
2010	1	http://mediafire...	3074
2010	1	http://linksynerg...	3037
2010	2	http://nfl.com/Gr...	4816
2010	2	http://eonline.co...	3609
2010	2	http://linkedin.c...	3519
2010	2	http://inbox.com/...	2880
2010	2	http://ebay.com/S...	2642
2010	2	http://shareasale...	2472
2010	2	http://wellsfargo...	2433
2010	2	http://outbrain.c...	2380
2010	2	http://theblaze.c...	2371
2010	2	http://mediafire...	2362
2010	3	http://nfl.com/Gr...	6427
2010	3	http://linkedin.c...	4411
2010	3	http://eonline.co...	4358
2010	3	http://inbox.com/...	3765

Fig 6: url count by all users during each month

C. Employee details:

This is the dataset which provides the details of the employee like the name, e-mail, role, supervisor, business unit, etc. We have concatenated the all the 12 employee detail files into one for easy processing. With the outcomes from device info and http info files and mapped these three tables by user so that it will give a clear picture of user behavior.

user_id	employee_name	email	role	supervisor	spam_count	average_duration_per_day
CM00912	Caleb Mason Ball	Caleb.Mason.Ball@...	Manager	Todd Vance Spears	0	12.64
SA00781	Serina Ariana Butler	Serina.Ariana.But...	Technician	Blaine Helen Cerv...	0	10.79
JO00169	Jonah Dakota Bird	Jonah.Dakota.Bird...	Salesman	Meredith Ainsley ...	0	10.55
RI00732	Risa Hillary Obrien	Risa.Hillary.Obri...	Salesman	Lysandra Chastity...	0	11.69
CM00912	Caleb Mason Ball	Caleb.Mason.Ball@...	Manager	Todd Vance Spears	0	12.64
SA00781	Serina Ariana Butler	Serina.Ariana.But...	Technician	Blaine Helen Cerv...	0	10.79
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CM00912	Caleb Mason Ball	Caleb.Mason.Ball@...	Manager	Todd Vance Spears	0	12.64
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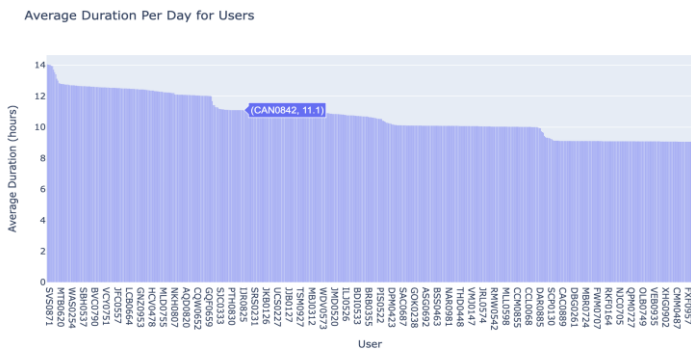


Fig 7: Average working duration per user

D.Files.csv:

The exploratory data analysis (EDA) was performed on a CSV file containing 2,500,000 rows and five columns: user_id, pc, time, filename, and content. This file captured information about user interactions with various files within the system.

During the EDA process, we analyzed the distribution of file accesses across different hours of the day. By aggregating the data based on the time column, we could visualize the hourly patterns of file access activity. This analysis revealed insights into peak usage hours, which could inform resource allocation and system optimization strategies.

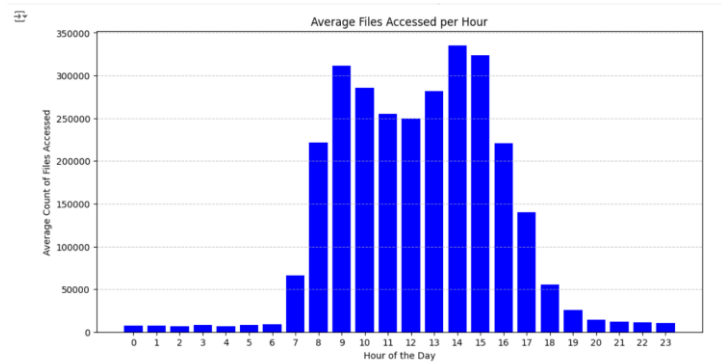


Fig 8: Average Files Accessed per Hour

Additionally, we examined the file access count by role, leveraging the user_id column to identify distinct user roles or groups. By aggregating the data based on user roles, we could understand the file access patterns and requirements of different user segments. This information could be valuable for tailoring access permissions, prioritizing file management efforts, or identifying potential bottlenecks or security concerns specific to certain user roles.

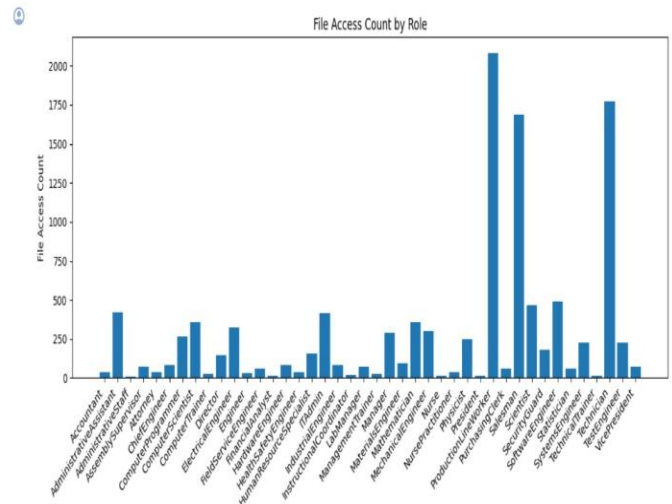


Fig 9: File Access count by role

The EDA process provided valuable insights into the file access patterns within the system, enabling data-driven decisions for performance optimization, resource allocation, and user experience enhancement.

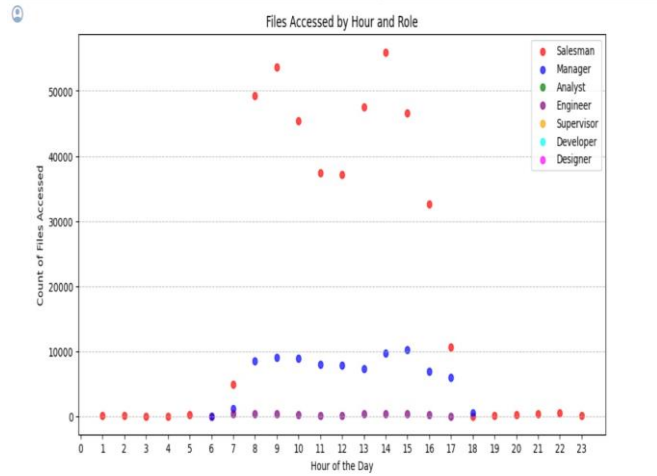


Fig 10: File accessing pattern by hours

V. MODELLING

A. Classifying Malicious URL's:

For classifying whether an URL is an malicious or not first we need to define or identify how/what is an malicious url is. So basically malicious urls has different characteristics like url length, TF-IDF feature of url, and sentiment score of the content etc.,. And then we can define a set of words which looks like offer, scan, phishing, discount, etc., and check whether they are present in any of the URL and if so, what is the frequency. So instead of checking whether these words are presented in the entire dataset as the data is pretty large i.e. 10M rows, we try to check them on more likely malicious URLs. How can we say a set of URLs as more likely malicious, if we can separate the whole URLs based on the features like URL length TF-IDF of URL and sentiment score we can find out the outliers using Isolation Forest, these outliers can be more likely malicious.

5.1 Sentiment Score:

I performed sentiment analysis was performed using the VADER (Valence Aware Dictionary and sEntiment Reasoner) model from the NLTK (Natural Language Toolkit) library. VADER is a well-regarded sentiment analysis tool that is specifically tuned to sentiments expressed in social media [10]. It provides a compound score, which is a normalized, weighted composite score that ranges from -1 (most extreme negative) to +1 (most extreme positive). To apply this model, the URLs accessed by employees were split into individual words, and a TF-IDF (Term Frequency-Inverse Document Frequency) vectorization was performed to quantify the importance of each word

in the context of the document set. The sentiment scores were then calculated for each URL content using the VADER model, which yielded a detailed sentiment profile of the accessed content. The analysis revealed significant variations in sentiment scores across different URLs, indicating diverse user interactions ranging from highly positive to highly negative content. This sentiment information was then used as a feature in the isolation forest model to detect outliers, helping to identify potentially malicious URLs. The integration of sentiment scores significantly enhanced the detection process, making it more robust and context-aware.

url	date	user	url_length	content	features	sentiment_score
http://esn.com/zh...	01-02-2010	06:55	LR00148	75 remain representa...	(262144, [24698, 55...	0.8519
http://webnspoon...	01-02-2010	07:00	NGF0157	119 festival off nort...	(262144, [11990, 19...	0.4939
http://aa.com/Rho...	01-02-2010	07:03	NGF0157	74 long away reorgan...	(262144, [9129, 127...	0.6124
http://groupp.co...	01-02-2010	07:05	IRW0931	80 among german schw...	(262144, [2931, 538...	0.3812
http://flickr.com...	01-02-2010	07:05	IRW0931	102 kate criteria j 2...	(262144, [1652, 122...	0.6486
http://skype.com...	01-02-2010	07:05	IRW0931	93 feed commonly ree...	(262144, [7155, 127...	-0.296
http://wikipedia...	01-02-2010	07:06	LR00148	102 programs must lon...	(262144, [6099, 136...	0.5574
http://constantco...	01-02-2010	07:06	IRW0931	121 columns ceased di...	(262144, [6498, 994...	-0.4588
http://tribalfusi...	01-02-2010	07:07	NGF0157	92 newly laying earn...	(262144, [3564, 655...	0.73
http://newworld.co...	01-02-2010	07:07	LR00148	97 himsel' delighted...	(262144, [2366, 649...	0.7651
http://sdmds.co...	01-02-2010	07:08	IRW0931	75 october 1 228 pus...	(262144, [1272, 232...	-0.8352
http://gap.com/Ak...	01-02-2010	07:08	NGF0157	92 refuses plus bast...	(262144, [1765, 642...	0.4019
http://files.word...	01-02-2010	07:11	LR00148	96 performance 5 rev...	(262144, [2366, 256...	0.7668
http://wikipedia...	01-02-2010	07:12	LR00148	102 1989 3 san master...	(262144, [1303, 154...	-0.2144
http://sfgate.com...	01-02-2010	07:12	LAP0338	90 remnant red front...	(262144, [1921, 500...	0.7717
http://sfgate.com...	01-02-2010	07:12	LAP0338	90 depth family nucl...	(262144, [1525, 495...	0.2732
http://sfgate.com...	01-02-2010	07:12	LAP0338	90 rising college se...	(262144, [8804, 100...	0.3182
http://w.com/92...	01-02-2010	07:13	LAP0338	87 thereby located i...	(262144, [12191, 15...	-0.0534
http://amazonwe...	01-02-2010	07:13	LAP0338	111 insufficient nept...	(262144, [1164, 183...	-0.8355
http://yfong.com/...	01-02-2010	07:13	LAP0338	63 rings humidity pa...	(262144, [11422, 11...	-0.128

Fig 11: Sentiment score for the file content

5.2 Isolation Forest :

The Isolation Forest algorithm was employed in this project to detect anomalous behavior in employee activity. Isolation Forest is an unsupervised machine learning algorithm specifically designed for anomaly detection, which operates by isolating observations through randomly selected attributes and splitting values [11]. In our implementation, we used a combination of features including the TF-IDF scores of the URLs accessed by employees, the length of the URLs, and the sentiment scores derived from the VADER model. These features were combined into a single feature vector for each URL using the VectorAssembler in PySpark. The Isolation Forest algorithm then processed these feature vectors to identify outliers, which are instances that appear less frequently and are isolated quickly in the random partitioning process. This method effectively highlighted URLs that deviated significantly from typical usage patterns, indicating potential malicious activity. By leveraging these diverse features, the Isolation Forest algorithm provided a nuanced and comprehensive approach to anomaly detection, enhancing our ability to flag suspicious URLs and mitigate security threats.

```

from sklearn.ensemble import IsolationForest
from pyspark.ml.feature import VectorAssembler
from pyspark.sql.functions import col
from pyspark.sql.types import DoubleType

transformed_df = transformed_df.withColumn("sentiment_score", col("sentiment_score").cast(DoubleType()))

# Create a VectorAssembler for the features and sentiment score
assembler = VectorAssembler(
    inputCols=["features", "sentiment_score", "url_length"],
    outputCol="features_combined"
)

transformed_df = assembler.transform(transformed_df)
# Convert to pandas DataFrame for scikit-learn
pandas_df = transformed_df.select("features_combined").toPandas()
# Train the Isolation Forest model
isolation_forest = IsolationForest(contamination=0.01)
predictions = isolation_forest.fit_predict(pandas_df[["features_combined"]].values.reshape(-1, 1))
# Add the predictions to the dataframe
transformed_df = transformed_df.withColumn("prediction", col(predictions))

# Show the predicted outliers
transformed_df.select("url", "content", "features", "sentiment_score", "url_length", "prediction").show(truncate=False)

```

Fig 12: Performing Isolation Forest

Note: This sentiment score is used as one of the features for isolating the probable malicious URL's. Thus, the process of finding the outliers in the dataset doesn't solely rely on this sentiment score, its just one of the features.

5.3 Classification:

Now its time to classify whether a URL is an malicious or not, but this looks simple as we only have a subset of data and we can easily classify/ identify from those. We have used an pre-trained ml model(Random forest Classifier) trained on comprehensive dataset from various datasets available publicly. It just forms the features like splitting the url into wrds , removing stop words and checking the format of the url like IPv4 or IPv6 etc.. and it performed well on test dataset i.e the accuracy was 94%. I have used the same model on our isolatd subset of url data and tried to check presence of malicious URL's.

B. Recommender System:

In the recommender system, a content-based approach is used to recommend files to users. Initially, the cosine similarity between files is calculated to measure the similarity in their content. Since the dataset contains a large number of unique files, the resulting similarity table becomes large (nxn), posing challenges in performing actions on it efficiently.

To address this, K-means clustering is applied to cluster files based on their content IDF (Inverse Document Frequency) scores. This clustering step helps in grouping similar files together, reducing the complexity of the similarity matrix. The files are clustered into 5 clusters, providing a structured way to recommend files based on their content similarities.

For recommendation, if a user belongs to a particular cluster but hasn't opened any files in that cluster, the system recommends the top 5 files from that cluster to the user. This approach ensures that users receive personalized recommendations based on the content they are likely to be interested in, improving the overall user experience and engagement with the platform.

filename1	filename2	similarity
EYPC9Y08.doc	N3LTSU30.pdf	0.7601627412030295
EYPC9Y08.doc	QCSW62YS.doc	0.6320646124193537
EYPC9Y08.doc	J509VZ0J.doc	0.8253719689398838
EYPC9Y08.doc	M3VNF4DQ.doc	0.6588500465028247
EYPC9Y08.doc	LX0I6B1U.pdf	0.7550904070248406
EYPC9Y08.doc	WX1IWCEK.txt	0.7225707224959798
EYPC9Y08.doc	IDJDLCL3.pdf	0.7928117742078699
EYPC9Y08.doc	THZZWCOX.doc	0.8356848301456895
EYPC9Y08.doc	STZG8JBH.pdf	0.7409651735034605
EYPC9Y08.doc	VX3DF6HY.doc	0.7626630302340263
EYPC9Y08.doc	RGVO2S20.zip	0.77156552713942
EYPC9Y08.doc	TMF79YY1.doc	0.6720051240543955
EYPC9Y08.doc	QCBWUV80.txt	0.7558424460304205
EYPC9Y08.doc	FVLT1244.txt	0.6451399333625208
EYPC9Y08.doc	X358UQDL.pdf	0.7479763673574423
EYPC9Y08.doc	G8I6GFRO.doc	0.6137242175311584
EYPC9Y08.doc	OYAT3WVN.zip	0.7111214986647387
EYPC9Y08.doc	OU7QK1UC.doc	0.7173220317953649
EYPC9Y08.doc	T08YFSTM.pdf	0.7016646115732518
EYPC9Y08.doc	OMIOJHFF.doc	0.6347882369642728

only showing top 20 rows

Fig 13: Cosine Similarity between files

```
User: HAD0246, Recommended Files: ['1P060RXU.doc', 'V3Z7LPHR.txt', 'ZG6E1J50.pdf', 'I58R00N1.txt', 'Y89KYD7D.txt']
User: TDS0984, Recommended Files: ['1P060RXU.doc', 'V3Z7LPHR.txt', 'ZG6E1J50.pdf', 'I58R00N1.txt', 'Y89KYD7D.txt']
User: IWS0280, Recommended Files: ['STZG8JBH.pdf', 'YV6ABHLE.doc', 'DS0NINR7.doc', 'K1WANDSV.doc', '8HSRQ0ZS.doc']
User: MPW0220, Recommended Files: ['3QKY90BK.txt', 'KTQNUG60.doc', 'R3M1FAGT.doc', 'OHQVUZTO.doc', 'I26NV6KY.doc']
User: BDT0533, Recommended Files: ['STZG8JBH.pdf', 'YV6ABHLE.doc', 'DS0NINR7.doc', 'K1WANDSV.doc', '8HSRQ0ZS.doc']
User: BAJ0654, Recommended Files: ['1P060RXU.doc', 'V3Z7LPHR.txt', 'ZG6E1J50.pdf', 'I58R00N1.txt', 'Y89KYD7D.txt']
```

Fig 14 : Files Recommendation per user

In this way we were able to recommend top 5 similar files to corresponding users which he/she haven't accessed them.

C. Creating User Profiles:

In order to identify the user-user relation and patterns we need to know their activities and their relation. Here we tried to find few features like average hours worked by a user per day, what is the peak time that user likely to work, this is calculate by the finding the count file and URLs accessed by a user at across the day and finding the max out of it. And we also found the first and last login date so that we can understand the relation of a user with the company. By analyzing all these findings, we can form a meaningful relational table which shows the interactions between users.

user	day_of_week	login_count	first_login_date	last_login_date	hour	activity_count
CMB0912	6	131	2010-01-03	2010-10-05	7	271
CMB0912	3	134	2010-01-03	2010-10-05	7	271
CMB0912	7	47	2010-01-03	2010-10-05	7	271
CMB0912	1	68	2010-01-03	2010-10-05	7	271
CMB0912	2	126	2010-01-03	2010-10-05	7	271
CMB0912	4	129	2010-01-03	2010-10-05	7	271
CMB0912	5	129	2010-01-03	2010-10-05	7	271
JDB0169	2	101	2010-01-04	2010-10-05	7	193
JDB0169	5	109	2010-01-04	2010-10-05	7	193
JDB0169	3	111	2010-01-04	2010-10-05	7	193
JDB0169	6	106	2010-01-04	2010-10-05	7	193
JDB0169	4	108	2010-01-04	2010-10-05	7	193
RHO0732	5	157	2010-01-04	2010-10-05	8	133
RHO0732	2	150	2010-01-04	2010-10-05	8	133
RHO0732	7	7	2010-01-04	2010-10-05	8	133
RHO0732	3	160	2010-01-04	2010-10-05	8	133
RHO0732	6	155	2010-01-04	2010-10-05	8	133
RHO0732	4	144	2010-01-04	2010-10-05	8	133
SAB0781	2	97	2010-01-04	2010-10-05	8	193
SAB0781	5	108	2010-01-04	2010-10-05	8	193

only showing top 20 rows

Fig 15: Tabulating user interactions.

D. Power BI Dashboards:

In our project, we developed a comprehensive Power BI dashboard to visualize and analyze the data collected on employee behavior and URL access patterns. The dashboard offers an in-depth view of the entire company's operations, including department-specific metrics such as the total hours worked and the top 5 URLs accessed by each department. Additionally, it provides detailed individual user analysis, showcasing average hours worked, URLs accessed, and personalized file recommendations. This interactive dashboard serves as a one-stop solution for all the analyses performed during the project, featuring user-friendly visuals and interactive elements that facilitate a deeper understanding of the data. By consolidating various data points and insights into a single platform, our Power BI dashboard enhances decision-making processes and provides actionable intelligence for improving security and productivity within the organization

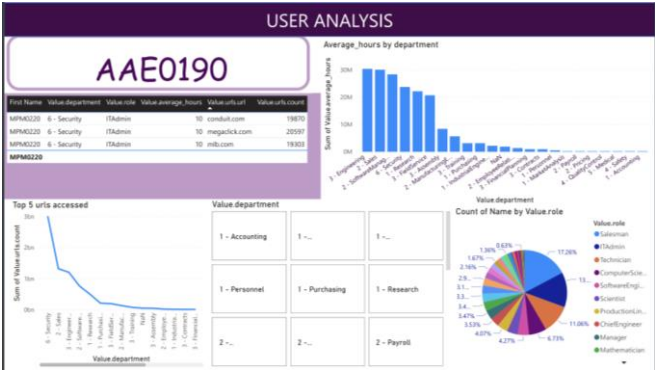


Fig 16: Final Dashboard using PowerBI

The above fig is an dashboard which shows all the basic statistics of an entire organisation across the year which includes what is the number of work hours contrinuted by an department, most accessing URL and number of employees by roles and their distribution

VI. DISCSSIONS & CONCLSIONS

A. Decisions Made/Things That Worked

In this project, we made several key decisions that contributed to its overall success. One of the primary decisions was to use a combination of machine learning algorithms for different tasks. For instance, we employed the Isolation Forest algorithm for anomaly detection to identify potentially malicious URLs based on employee behavior patterns. This choice proved effective due to the algorithm's robustness in handling high-dimensional data and its ability to isolate anomalies with minimal assumptions about the data distribution. Additionally, the implementation of a content-based recommender system using cosine similarity and K-means clustering enabled us to provide personalized file recommendations for users based on their historical access patterns. This approach effectively grouped similar content and users, enhancing the relevance of recommendations.

Another successful decision was the integration of a sentiment analysis model using the VADER lexicon from the NLTK library. This model allowed us to assign sentiment scores to URLs accessed by employees, providing valuable insights into the nature of the content being accessed. Finally, the development of an interactive Power BI dashboard was instrumental in visualizing and analyzing the data, offering both company-wide and individual user insights in an accessible and user-friendly manner.

B. Difficulties Faced/Things That Didn't Work Well

Despite these successes, we encountered several challenges during the project. One significant difficulty was handling the large volume of data when calculating cosine similarity for the content-based recommender system. The resultant similarity matrix was massive, making it challenging to perform computations efficiently. To address this, we implemented clustering techniques to reduce the data's dimensionality, but this approach had its limitations in terms of precision.

Another challenge was the initial performance of the Isolation Forest algorithm. Tuning the model's hyperparameters to balance sensitivity and specificity required extensive experimentation and validation against various datasets. Additionally, integrating different data sources and ensuring data quality and consistency were ongoing challenges that required meticulous preprocessing and validation efforts.

B. Conclusions:

Overall, the project successfully achieved its primary objectives of analyzing employee behavior patterns and developing a recommender system to suggest relevant files to users. By leveraging a combination of machine learning algorithms for anomaly detection and content recommendation, we were able to provide valuable insights and enhance security measures within the organization. Throughout the project, we learned the importance of integrating various data sources and processing large datasets efficiently using PySpark, which allowed us to handle and analyze vast amounts of data seamlessly.

Our findings revealed several key patterns in employee behavior, including typical and atypical login/logout activities and URL access trends. These insights enabled us to identify potential security threats early and take proactive measures to mitigate them. The implementation of the Isolation Forest algorithm for anomaly detection proved effective in filtering out malicious URLs, while the Random Forest model demonstrated high accuracy in classifying URLs based on a mix of features, such as TF-IDF scores and sentiment analysis results.

The Power BI dashboard served as a comprehensive tool for visualizing data, offering an interactive and user-friendly interface that facilitated decision-making and proactive security management. It provided detailed analyses at both the department and individual levels, including metrics like average hours worked, top URLs

accessed, and personalized file recommendations. This holistic view of employee behavior and access patterns allowed for more informed and strategic security decisions.

In conclusion, the project not only met its goals but also provided a robust framework for ongoing security and productivity enhancements. Moving forward, there is significant potential for further improvements and expansions to enhance the system's capabilities and accuracy. These include refining the anomaly detection models, incorporating more advanced machine learning techniques, and continuously updating the recommender system with new data to ensure its relevance and effectiveness.

VI. WORK DISTRIBUTION

We are group of two Evakattu Muni Eshwar and Pallavi Suma working on this project. We have divided the task as per the use cases, where she worked on how many hours a user worked per day, to achieve this she used `login_info` and employee details datasets and building an impressive dashboard using PowerBI. I have worked on spam/ham URL classification using unsupervised learning and building a recommender system for files.csv. Working on these use cases include data handling, preprocessing, data cleaning, feature engineering ml modeling on our respective datasets.

VII. FUTURE SCOPE

The future scope for this project involves expanding its capabilities and improving its accuracy and efficiency. One potential direction is the integration of real-time data processing to enable immediate detection and response to anomalous behavior and potential security threats. Additionally, incorporating more advanced machine learning algorithms, such as deep learning models, could enhance the accuracy of behavior analysis and URL classification. Another area for expansion is the development of more sophisticated recommendation systems that leverage hybrid approaches, combining content-based and collaborative filtering techniques to provide even more personalized and relevant file recommendations. Moreover, the system could be extended to analyze additional data sources, such as email and messaging patterns, to provide a more comprehensive view of employee behavior. Finally, enhancing the Power BI dashboard with more advanced interactive features and predictive analytics capabilities could further support decision-making processes and proactive security measures within the organization.

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